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Understanding the burden of depression, anxiety and stress among first-year undergraduate students

Umar Haruna¹ , Abdul-Raheem Mohammed^{1*} and Mary Braimah²

Abstract

This study investigated the levels of depression, anxiety, and stress among first-year university students and predictors of depression, anxiety, and stress among this cohort of students. Six hundred and eighty-eight (688) first-year undergraduate students were sampled for the study. The Depression, Anxiety, and Stress Scale-21 (DASS-21) was used to assess the prevalence of mental health issues. The results showed prevalence rates for depression, anxiety, and stress among students were 49%, 75%, and 73.26%, respectively. Females reported higher levels of all three variables. There were positive correlations between academic stress and depression ($r=.27$), anxiety ($r=.21$), and stress ($r=.23$), all significant at $p<.001$. These results suggest that depression, anxiety, and stress are common mental health issues among undergraduate students, affecting individuals to varying degrees and influenced by a range of factors.

Keywords Stress, Anxiety, Depression, Undergraduate students, Prevalence, Mental health

Introduction

Mental health issues such as anxiety, depression, and stress are among the foremost global health concerns [1]. Approximately 6% of adults worldwide suffer from Major Depressive Disorder, and 4–7% suffer from Generalised Anxiety Disorder [2]. Depression and anxiety have recorded prevalence rates of 9% and 10%, respectively, among the general population [3]. Anxiety disorder is considered among the most prevalent mental health disorders [4]. Prolonged stress exposures result in depressed conditions [5, 6]. A popular tool that is often used to explore the presence of depression, anxiety, and

stress in the general population is the DASS-21. The DASS-21 posits that while depression, anxiety, and stress are all part of a broader general distress construct, each also exhibits unique features. Depression is marked by a diminished experience of positive emotions, anxiety is associated with heightened physiological arousal, and stress is characterised by symptoms such as irritability, nervous tension, trouble relaxing, and agitation [7, 8].

These disorders affect people's health and quality of life [9]. For instance, depression impacts various aspects of a person's life, such as daily functioning, mood, interests, appetite, cognition, and energy levels [10, 11]. Anxiety negatively impacts concentration, sleep, mood, and academic performance [12]. Stress often leads to negative consequences such as increased smoking, substance use, accidents, sleep problems, and eating disorders [13]. Despite the overwhelming negative impact of stress, anxiety and depression, less attention has been devoted to their prevalence and predictors among first-year

*Correspondence:

Abdul-Raheem Mohammed
arm.raheem@gmail.com

¹Department of Social and Behavioural Change, School of Public Health, University for Development Studies, Tamale, Ghana

²Department of Guidance and Counselling, University for Development Studies, Tamale, Ghana



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university students in sub-Saharan Africa. The study aims to examine the correlates and predictors of stress, anxiety and depression among first-year university students.

University is considered a transitional period for young people. It is a time of adjustment to a new institution and new peers for young adults. This transition can impact the mental well-being of young people, who appear to experience more psychological challenges compared to their peers who do not attend university and the general population [14]. University students often face a mix of physical, psychological, social, and academic challenges, making them more susceptible to psychological distress such as depression, anxiety, and stress [15]. This is more challenging in sub-Saharan Africa, where general living standards are low and lower levels of income, which affects quality of life among these students.

According to Erikson's psychosocial developmental stages [16, 17], university students are in the stage of exploring and finding their own identity. They are trying to understand who they are and what they want to become, and are also learning to become independent from their parents or caregivers. Young people, such as university students, are at a stage where they need to learn how to interact and connect with others. These changes, such as exploring career options, moving away from home, and learning to interact with new people, often contribute to the stress and anxiety experienced by university students.

Brook and Willoughby [18] observed that while many university students successfully navigate these developmental stages, adjusting academically and psychosocially is a difficult issue for many others. University students meet new people daily, endure increased academic obligations from both lecturers and parents and have increased exposure to things like alcohol and or drugs. These experiences can affect and alter their identity by changing values, attitudes, and goals [18]. Due to the numerous challenges university students experience, they are at a high risk of mental health problems such as depression, anxiety and stress. Specifically, they experience a unique type of stress related to the academic environment is academic stress. Academic stress includes key components such as academic expectations, workload and exams, and academic self-perception [19]. It has been found that academic stress has an impact on overall stress, including elevated stress cortisol and perceived stress [20].

Stress and anxiety are normal daily experiences for university students because they are exposed to different stressors, notably, the pressures from academic tests, difficulties associated with integrating into the system and uncertainty of students' future [21, 22]. Depression, anxiety, and stress are important indicators of mental health, especially among university students [23].

Studies have asserted that an alarming number of students are unable to cope with university curriculum demands and parents' expectations, resulting in negative mental health concerns such as anxiety and depression for students [24]. Depression is common among university students and has far-reaching consequences, such as increased abuse of drugs, comorbid with mental health conditions like anxiety and suicide [25]. Liu et al. [26] report that studies conducted in various countries have revealed the following prevalence rates of depression among college students: 18.4% in Spain [27], 34% in Turkey [28], 34.5% in the UK [29], and 37.2% in Jordan [30].

It is reported that one out of every five young people has mental health problems, namely anxiety, depression, and stress, among others [31]. It has also been demonstrated that depression, anxiety and stress can be manifested in different forms; however, some common overt symptoms include loss of appetite, sleep disturbance, lack of concentration, apathy (lack of enthusiasm and concern), and poor hygiene [32].

Many undergraduate students start university at a young age, and dealing with depression, anxiety, and stress early in life can have long-term negative effects on students' mental and social well-being [33]. Nakie et al. [34] and Sandal et al. [35] also emphasise the importance of acknowledging university students' unique challenges, who often experience separation from family for the first time. This transition involves increased academic pressures, the need for self-sufficiency, and the formation of new social networks, all of which can affect their mental and emotional well-being.

The present study

Although some attempts have been made to explore the prevalence of stress, anxiety and depression from the sub-Saharan perspective, they largely focus on the entire student population. However, the first year of undergraduate education is a critical period that requires life adjustments for these students. Thus, a comprehensive understanding of these challenges is essential during this critical stage of their lives. Therefore, this study aims to (1) investigate the levels of depression, anxiety, and stress among university first-year students, (2) analyse the factors associated with these mental health issues, and identify predictors of depression, anxiety, and stress among this cohort of students. This study offers an important opportunity to advance the understanding of the prevalence and associated correlates of depression, anxiety, and stress among students in developing countries. This can help governments, universities, families, and health-care agencies identify and minimise risk factors associated with mental health problems.

Methods

Study design

A quantitative research design was utilised for the study, employing a cross-sectional survey methodology. Participants completed a series of questionnaires that gathered information on demographic characteristics, perceived levels of stress and depression, as well as factors associated with depression, anxiety, and stress within the campus environment. This approach offers several beneficial features, including the ability to gather data efficiently, providing a broad snapshot of the issues at hand, and enabling the identification of patterns and correlations among various factors.

Participants

A total of 688 first-year undergraduate students from the University for Development Studies, Ghana, took part in this research, participating in exchange for course credit. The sample size was determined a priori. This involved the use of G-Power software, with a power of 90%, a two-tailed test, and effect sizes reported in previous studies. Participants were selected using a convenience sampling method, ensuring a practical approach to gathering data. The average age of the students was 20.69 years, with a standard deviation of 2.38, indicating a fairly consistent age range among them. The gender distribution revealed that a significant majority, 75.3%, of the participants were female, showcasing a notable demographic aspect of the student body. The students were drawn from four programmes of study, including Public Health (486), Dietetics (23), Community Nutrition (102) and Pharmacy (77).

The study was conducted in strict adherence to ethical guidelines concerning human participants and received approval from the research ethics committee of the Department of Social and Behavioural Change, School of Public Health, University for Development Studies (approval number S-EA-585-23). All students provided written consent, ensuring their willingness to be part of this investigation. First-year university students were specifically targeted for sampling due to the distinctive and formative nature of their initial university experiences, which can differ greatly from those of upperclassmen.

Data collection instruments and procedure

The Depression, Anxiety, Stress, Scale [7] was used to assess the level of depression, anxiety and stress among the students. This scale is made up of 21 items with

options ranging from 0 to 3, where they indicate how they felt over the past week (e.g., “I was worried about situations in which I might panic and make a fool of myself”). Each subscale, such as depression, anxiety, and stress, had seven items, where the responses to each subscale were summed up to constitute the score for each subscale. The total score for each subscale was multiplied by two. The Cronbach’s α for the scale in the present study was 0.91. Further details on the scale, including the items, are deposited in the supplementary file.

Academic stress was assessed using the perceived academic stress scale [19]. This scale is made up of 18 statements (e.g., I think that my worry about examination is weakness of character) with 5 options ranging from strongly disagree to strongly agree. Cronbach’s α for the scale in the present study was 0.75. Participants completed the questionnaires via a secure Google form link that was shared with them. They spent between 5 and 10 min completing the questionnaire.

Data processing and analysis

Data cleaning was meticulously performed using Microsoft Excel to ensure accuracy and reliability in the dataset. Following this process, we utilised JASP (version 0.16.3.0) to conduct our statistical analyses. For both the primary scales and their respective subscales, we computed means and standard deviations to summarise our data effectively.

To assess the distribution of our data, we conducted tests for normality, which confirmed that our dataset met the normality assumption. As a result, we proceeded with parametric tests for the analyses. Specifically, we employed the Pearson Product-Moment Correlation to explore the associations between the variables and implemented linear regression analysis to investigate how well the independent variables predict the dependent variables. Independent t-tests and one-way ANOVA were used to assess mean differences between two or more groups. Where necessary, post-hoc tests were applied to assess the specific differences found as a result of using the ANOVA.

Results

Prevalence of stress, anxiety and depression

The first objective of this study was to determine the prevalence of Stress, Anxiety, and Depression among first-year university students. The DAS 21 scale was applied. The overall mean scores for stress, anxiety, and depression are presented in Table 1, with the corresponding proportion of the severity of the variables illustrated in Fig. 1. The distribution of the levels of stress, anxiety, and depression showed that 26.74%, 25%, and 51% of the respondents had normal levels, respectively. Collectively, the prevalence rate ranging from mild to severe forms for

Table 1 Descriptive statistics of the outcome variables

Scale	Subscale	Mean	SD
	Stress	15.6	9.25
	Anxiety	14.2	9.2
	Depression	15.1	9.83
Total DASS		44.9	25.7

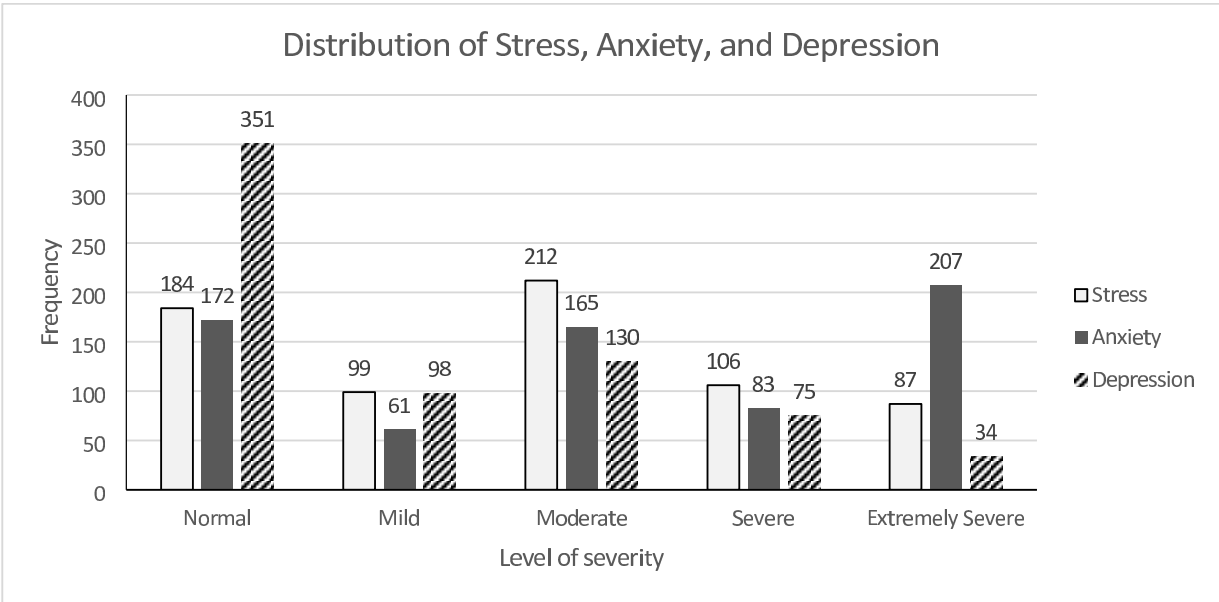


Fig. 1 Distribution of the levels of stress, anxiety and depression

Table 2 Summary of independent t-test for stress, anxiety, and depression based on gender and religion

Variable	Group	Mean	SD	SE	t	df	p
Stress	Female	16.2	9.24	0.4	3.08	686	0.002
	Male	13.7	9.03	0.69			
Anxiety	Female	15.0	9.21	0.41	4.07	686	< 0.001
	Male	11.7	8.77	0.67			
Depression	Female	15.8	9.74	0.43	3.12	686	0.002
	Male	13.1	9.86	0.76			
Total DAS	Female	47.0	25.55	1.12	3.76	686	< 0.001
	Male	38.5	25.32	1.94			
Stress	Christian	15.4	9.72	0.57	-0.45	686	0.65
	Muslim	15.7	8.91	0.45			
Anxiety	Christian	13.8	9.29	0.55	-0.95	686	0.34
	Muslim	14.5	9.14	0.46			
Depression	Christian	14.4	9.91	0.58	-1.54	686	0.13
	Muslim	15.6	9.76	0.49			
Total DAS	Christian	43.6	26.45	1.56	-1.09	686	0.28
	Muslim	45.8	25.2	1.26			

stress, anxiety, and depression among the students was 73.26%, 75%, and 49%, respectively.

Predictors and correlates of depression, anxiety and stress
The study further analysed the factors associated with depression, anxiety, and stress among the group. Three

demographic factors and two school-level factors were explored. These included age, gender, religion, program of study, and academic stress. The results of the independent t-test for gender and religion and the bivariate correlations are summarised in Tables 2 and 3, respectively.

Table 3 Summary of the bivariate correlations among the study variables

Variable	Subscale	Stress	Anxiety	Depression	Total_DASS
Academic stress		0.234***	0.21***	0.27***	0.262***
	AE	0.203***	0.202***	0.198***	0.221***
	W&E	0.137***	0.102**	0.17***	0.151***
	AS-P	0.192***	0.186***	0.241***	0.228***
Age		-0.044	-0.01	-0.025	-0.029

Note: AE = Academic expectation, W&E = Workload & exam, AS-P = Academic self-perception, DASS = Depression, Anxiety, Stress Scale, * $p < .05$, ** $p < .01$, *** $p < .001$

Table 4 Summary of the results of linear regression on stress, anxiety and depression

Variable		B	SE B	β	t	p
General Stress						
Academic stress		0.12	0.019	0.23	6.3	< 0.001
R	0.23					
R ²	0.06					
Adj. R ²	0.05					
F	39.70					< 0.001
Anxiety						
Academic stress		0.11	0.019	0.21	5.62	< 0.001
R	0.21					
R ²	0.04					
Adj. R ²	0.04					
F	31.59					< 0.001
Depression						
Academic stress		0.15	0.02	0.27	7.34	< 0.001
R	0.27					
R ²	0.07					
Adj. R ²	0.07					
F	53.86					< 0.001

Note: B = unstandardized coefficient, SE = Standard error, β = Standardized coefficient

The distribution based on gender showed that females reported higher levels of stress ($t(686) = 3.08$, $p = .002$), anxiety ($t(686) = 4.07$, $p < .001$), and depression ($t(686) = 3.12$, $p = .002$) than males. However, there were no significant differences between males and females on the total academic stress scale ($t(686) = 0.37$, $p = .71$).

There was a significant positive correlation between perceived academic stress and depression ($r(688) = 0.27$, $p < .001$), anxiety ($r(688) = 0.21$, $p < .001$), and stress ($r(688) = 0.23$, $p < .001$). The results of the linear regression, as illustrated in Table 4, showed that academic stress was a significant predictor of stress [$F(1, 686) = 39.7$, $p < .001$] accounting for 6% of the variance on stress ($R^2 = 0.06$, $R^2 \text{ adj} = 0.05$). It was also a significant predictor of anxiety among the students [$F(1, 686) = 31.6$, $p < .001$] accounting for 4% of the variance on stress ($R^2 = 0.04$, $R^2 \text{ adj} = 0.04$). Finally, academic stress was found to be a significant predictor of depression among the students [$F(1, 686) = 53.86$, $p < .001$] accounting for 7% of the variance on stress ($R^2 = 0.07$, $R^2 \text{ adj} = 0.07$).

Age, religion and programme of study

Age did not correlate with depression ($r(688) = -0.03$, $p = .52$), anxiety ($r(688) = -0.01$, $p = .79$), and stress

($r(688) = -0.04$, $p = .25$). Religion did not have a significant impact on the reported stress ($t(686) = 0.45$, $p = .65$), anxiety ($t(686) = 0.95$, $p = .34$) and depression ($t(686) = 1.54$, $p = .13$). Programme of study did not have a significant impact on stress ($F_{(3, 684)} = 1.367$, $p = .25$) and depression ($F_{(3, 684)} = 2.11$, $p = .098$), except anxiety ($F_{(3, 684)} = 3.24$, $p = .02$) where post hoc results showed Public health students reported significant higher levels of anxiety than Pharmacy students ($t = 1.66$, $p = .02$).

Discussion

The current study investigated the prevalence and correlates of stress, anxiety and depression among first-year undergraduate students. The results showed that about 35% of the sample exhibited normal thresholds for the three mental health variables. In comparison, 12.5% and 24.6% were said to have met the criteria for mild and moderate symptoms, respectively. The findings demonstrate that first-year students indeed exhibited signs of stress, anxiety and depression. It appears that overall, anxiety, which occurred among 75% of the sample, is the mental issue with the highest prevalence, closely followed by stress (73.26%) and finally depression, with a prevalence rate of 49.3%. These high prevalence rates are

consistent with other studies, such as Panda et al. [36], who reported a depression prevalence rate of 57.6%, an anxiety rate of 74%, and a stress rate of 57.7%. Astutik et al. [37] also found that a quarter (25%) of students experienced depression, while slightly over half (51.5%) showed signs of anxiety, and 38.9% faced challenges with stress. Additionally, a study conducted among nursing students in a Ghanaian university revealed that 58% of students met the clinical criteria for depression, 58.4% for anxiety, and 65% for stress [23]. The higher prevalence of anxiety and stress among first-year students is not surprising. This group is not only transitioning into adulthood, as reflected in their age range, but they are also encountering new levels of independence during this time. The first year of university is a time that demands significant life adjustments, including adapting to lectures, preparing for examinations, navigating relationships, integrating into the university culture, and dealing with uncertainties about the future [21]. This is underscored by results indicating that academic stress contributes significantly to the overall stress experienced by students. This finding can be further understood through Erikson's theory of psychosocial development regarding this stage of life. According to Erikson, these students are in the developmental phase of self-identity, during which they explore who they are, and what they want to become in the future and learn to become independent from their caregivers [14, 15].

In terms of the severity of depression, anxiety, and stress, the findings are consistent with some of the existing literature. The research revealed a higher occurrence of depression, anxiety, and stress symptoms when compared to earlier studies [15, 38, 39, 40]. For instance, Lee et al. [38] indicated that students faced moderate to extremely severe levels of depression (36%), anxiety (45%), and stress (63%). Although the majority of these studies were conducted during the COVID-19 pandemic, the higher prevalence rates observed in the present study suggest that mental health challenges extend beyond health crises. They are also a significant concern for individuals adjusting to new environments, as seen among first-year undergraduate students. In the context of Ghana, Wireku-Gyebi [41] found that among final-year students, the prevalence of moderate to extremely severe depression was 67.9%, while moderate to extremely severe anxiety affected 76.6%, and 64.3% reported moderate to severe levels of stress. The different target groups of respondents may be responsible for these differences.

In terms of the correlates of depression, anxiety, and stress, gender appeared to have a significant association. Females report higher levels of stress, anxiety and depression than males. This is consistent with the literature on gender differences in mental health [42, 43, 44, 45]. However, previous research on gender differences in mental

health disorders among University students shows conflicting results. For example, Ochnik et al. [43] found that adult female University students experience higher levels of mental health problems than their male counterparts. In contrast, Jiang et al. [46] as well as Kugbey et al. [47] found that male students reported higher levels of depression, anxiety, and stress. Interestingly, Othman et al. [42] failed to find any gender differences in the mental health status of students. Recently, Diez et al. [44] found that in terms of gender differences, women generally had higher levels of depression, anxiety, and stress than men and were also overrepresented in severe depression, anxiety, and stress. However, in sharp contrast to prior research, Campbell et al. [48] did not identify any link between female gender and poor mental health.

Historically, women have been viewed as more inclined to report physical and somatic symptoms compared to men [49]. While this might suggest they genuinely experience more internal symptoms, it could also stem from social desirability bias, as women may be less hesitant than men to discuss their distress [44]. Female university students indicated greater concern for others and found it more challenging to concentrate on their academic work [50]. The observed trend in the current research may be partly explained by the tendency of females to show greater emotional expressiveness compared to males [51]. This heightened emotionality could make them more vulnerable to psychological issues such as depression, anxiety, and stress. Regarding gender, females may experience greater anxiety than males, and males potentially underreport their anxiety levels [52, 53].

Academic stress, which includes stress associated with workload and exams, and academic expectations, was a predictor of depression, anxiety and general stress among the students. This meant that the higher perception of workload and exams, including the expectations the students had of their academic journey, largely predicted their levels of depression, anxiety, and general stress. Specifically, when academic stress increases by one unit, depression, anxiety, and general stress increased by 27%, 21%, and 23% respectively. This finding is consistent with previous studies [5, 6] on the contribution of academic stress to depression, anxiety, and stress.

One of the key factors contributing to depression among first-year university students is the intense academic pressure they encounter. Many students arrive at university with high expectations based on their prior academic achievements. This transition to a more demanding academic environment frequently results in feelings of inadequacy, anxiety, and depression when they cannot meet these self-imposed standards. The pervasive fear of failure, along with the competitive atmosphere of university life, only serves to amplify these emotional struggles. It is crucial to acknowledge and address these

challenges to ensure students can thrive in their new environment. Efforts aimed at a significant reduction of depression, anxiety, and general stress among these first-year students must include interventions that would minimise the level of academic stress among these students.

Moreover, the findings reveal that in line with some earlier findings [48], the religion and age of respondents did not have any significant association with the level of prevalence of depression, anxiety, and stress. A recent study that focused on high school students who are relatively younger than University students reported high prevalence rates of anxiety and depression [54]. With regards to the relationship between the programme of study and mental health, the study found a partial association. This resonates with previous studies that have focused on medical students [39], student nurses [23], and students studying hospitality courses [41] among others.

The findings present important implications for practice. The study reveals the incidence of depression, anxiety, and stress among first-year undergraduate students in Ghana, emphasising that mental health issues remain a significant public health concern. University students frequently encounter various challenges and stressors that may contribute to mental health issues like depression, anxiety, and stress. Understanding the underlying correlates of these problems is crucial for designing effective interventions and support systems. It was evident that depression, anxiety, and stress were common among undergraduate students, with varying levels of severity. Overall, stress and anxiety were more prevalent among these students, while depression occurred less frequently. Gender, type of program, and academic stress were identified as significant factors associated with these mental health issues. These findings emphasise the necessity of addressing the mental health needs of this vital group of potential human resources. To promote student well-being, educational institutions, healthcare organisations, and policymakers must take proactive measures to support their mental health. To address the incidence of depression, anxiety and stress, Universities should implement some practical measures. Interventions should focus on increasing support for female students while also de-stigmatising help-seeking behaviours among male students.

Institutions can maximise the use of social support systems, such as counselling units and academic tutoring, to provide essential counselling and support to students, especially those in their first year. However, it should be noted that the level of overall mental health problems in universities has the potential to create a huge burden on university counselling services to meet students' demands. Hence, more novel approaches, such as online

counselling services, can help universities meet increased demands.

Existing orientation sessions for freshmen and women should focus on mental health and preparing for university life to offer the necessary support. Students in various academic years face distinct risk factors from the moment they start university until they complete their studies; thus, students at different levels need specific coping strategies. Universities need to recognize these risk factors and take steps to reduce them while offering mental health support to their students. Furthermore, a programme such as Mental Awareness Week could be created to enhance mental health education among students. This could involve, mental health screening and assessments, faculty training on mental health support, mobile mental health-focused podcasts, counselling centre mental health websites and resources, stress management and coping skills programs, Time management and goal setting seminars, Academic support programs such as study skills seminars, research project support groups, peer mentoring groups.

Limitations and conclusion

This study has some limitations that should be acknowledged. First, the sample was predominantly female, which may limit the generalizability of the findings across genders. Interpretations of the results should therefore be made with caution, particularly in relation to male populations. Second, the use of a cross-sectional survey design and self-report measures restricts the ability to infer causality among the variables studied. Third, the use of convenience sampling, a non-probabilistic method, may limit the generalizability of the findings to the broader population. However, to reduce potential selection bias, participants were systematically recruited from a range of academic programs. This strategy helped to enhance the representativeness of the sample and minimise the risk of overrepresenting any particular subgroup. We failed to clarify if some of the students were already on treatment for any mental health disorder, although we did indicate that if any student has a mental health issue and needs some assistance, they could contact the principal investigator. Future research is needed to explore the long-term effects of depression, anxiety and stress on students. A longitudinal study with a large, randomly recruited sample that encompasses various age groups, genders, fields of study, and socioeconomic statuses is essential to examine how students' mental health evolves from their entry into university until graduation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12888-025-07069-8>.

Supplementary Material 1

Author contributions

U.H: Conceptualization, Methodology, Writing- Original draft preparation, Writing- Reviewing and Editing. A.-R.M: Conceptualization, Methodology, Formal analysis, Writing- Original draft preparation, Writing- Reviewing and Editing. M.B: Methodology, Writing- Original draft preparation, Writing- Reviewing and Editing.

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Data availability

The datasets generated by the survey research during and/or analyzed during the current study are available in the Open Science Foundation repository, https://osf.io/zt8kd/?view_only=96019554c58541a1b1706ba3beb6c368.

Declarations**Ethics approval and consent to participate**

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics committee of the Department of Social and Behavioural Change, School of Public Health, University for Development Studies (approval number S-EA-585-23) on August 15, 2023.

Consent to participate

All participants provided written informed consent prior to participating in the study.

Consent for publication

All participants consented to publication of the research output ensuring anonymity of the participants. All authors also approved the final version of the manuscript for publication.

Competing interests

The authors declare no competing interests.

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